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LETTER

The Coordinate-Space Geometry of the Tetrad Frame

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Abstract
The tetrad (vierbein) is the local orthonormal frame on which the coupling of gravity to fermionic matter depends, yet its coordinate-space geometry has not, to our knowledge, been examined. Deriving four oriented half-space inequalities from the spatial sector of a single orthonormal tetrad, the $2^4 = 16$ possible sign combinations yield exactly two bounded regions. The bounded pair form a stellated octahedron whose boundary decomposes into sixteen independent channels. No dual frame is introduced; the orientation-reversed tetrahedral sector emerges as the unique second bounded region of the same single-tetrad construction. All structural constants of the resulting geometry reduce to two integers: the number of tetrahedral faces $K = 4$ and the spatial dimension $D = 3$. A numerical rendering at four million sample points confirms the derived structure.

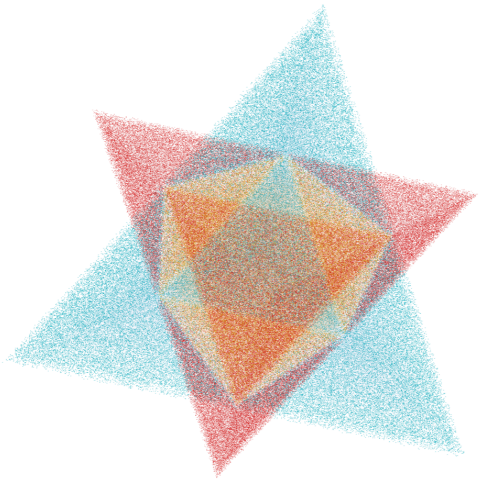


Figure 1. The coordinate-space geometry of the tetrad frame. Four million points are sampled uniformly in local frame coordinates and classified by the sign combination each point occupies under the four projections v_i. The primary tetrahedron (TetA) appears in blue, a second inverted tetrahedron (TetB) in red, and their shared interior, a regular octahedron, in gold. The geometry is analysed in the text. An interactive HTML rendering is provided as supplementary material.

Introduction

General relativity is most commonly formulated in terms of the metric tensor $g_{\mu\nu}$, a symmetric matrix encoding distances and angles in curved spacetime. There is, however, a deeper formulation. The *tetrad* (or *vierbein*) $e^a{}_\mu$ is a set of four vector fields that serve as a local orthonormal frame at each point of the manifold. The metric is recovered as a derived quantity,

$$g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu, \tag{1}$$

making the tetrad, in a precise algebraic sense, the square root of the metric. The flat metric η_{ab} in this expression carries the Lorentz signature $(-,+,+,+)$, one timelike dimension and three spacelike dimensions, and so the tetrad's four legs reflect this $3+1$ decomposition directly, one spanning the local time direction and three spanning the local spatial frame.

Coupling fermions (electrons, quarks, neutrinos) to gravity requires the local Lorentz frame that only the tetrad provides, making this formulation not an alternative to general relativity but its most fundamental expression. The approach has been standard since the work of Cartan, Einstein [1], and Weyl [2], and underpins modern frameworks including Poincaré gauge gravity [3], Ashtekar variables [4], loop quantum gravity, bimetric gravity [5], and supergravity.

Despite this centrality, the tetrad formalism is almost always presented in index notation: $e^a{}_\mu$ carrying a Lorentz index a and a spacetime index μ. The indices are manipulated algebraically, but in over a century of use, the spatial structure that these four vector fields generate has not, to our knowledge, been rendered in coordinate space.

This paper does that. We write the tetrad's spatial components as explicit half-space inequalities and derive the geometry that their half-space combinations produce; a numerical rendering confirms the result visually, but the argument is analytic.

We do not begin with a stellated octahedron, nor with a pair of dual tetrahedra. We begin with a single orthonormal tetrad. The opposite-oriented tetrahedral sector appears as a necessary consequence of the half-space structure that the tetrad's four planes generate.

Four Planes

The $(-,+,+,+)$ signature maps null directions in the local frame to the unit sphere S^2. The three identical spacelike entries give the spatial

sector full $SO(3)$ invariance; requiring that a discrete four-direction set on S^2 introduce no net preferred spatial direction uniquely fixes the regular tetrahedral frame with pairwise inner product $c = -1/3$ (proved in Supplementary Material, Proposition). The four directions, expressed as linear projections of the local coordinates, are:

$$\begin{aligned} v_1 &= x + y + z, \\ v_2 &= x - y - z, \\ v_3 &= -x + y - z, \\ v_4 &= -x - y + z. \end{aligned} \tag{2}$$

Each projection v_i defines a half-space: the set of all points on one side of the corresponding plane $v_i = \text{const}$. The four planes together close to form a tetrahedron, the unique bounded solid that four planes can enclose. The region where all four projections are simultaneously bounded defines the primary tetrahedral frame, which we label TetA:

$$\textbf{TetA}: \quad \max(v_1, v_2, v_3, v_4) \leq 1. \tag{3}$$

Each v_i is the dot product of the position vector with one of the four directions derived above; it measures the displacement of a point along that direction. These projections encode the tetrad's spatial sector in its unique isotropic four-direction basis, playing the role of spatial one-forms projected along the tetrahedral normals rather than the three orthogonal legs. Bounding them as half-space inequalities makes the tetrad's spatial geometry directly computable.

To render this geometry, we sample four million points uniformly in a cube of local frame coordinates, evaluate the four inequalities at each point, and colour each point according to the region it occupies. The geometry emerges from the half-space boundaries alone, without mesh construction or surface fitting. Figure 1 confirms the analytically derived structure described in the following section.

The Emergent Geometry

Figure 1 displays the geometry that the four half-space inequalities produce. The rendering reveals two tetrahedral regions, one upright (blue) and one inverted (red), overlapping around a compact gold core. All three regions emerge from the same four planes defined in the previous section. This section proves which regions must close into finite solids, why no others can, and identifies the composite solid they form.

Each projection v_i divides coordinate space with a bounding plane. On one side lies the half-space $\{v_i \leq 1\}$; on the other, its complement. Replacing v_i with $-v_i$ and applying the same bound gives the opposite half-space $\{-v_i \leq 1\}$. Since there are four projections, each offering two choices, there are $2^4 = 16$ ways to select one half-space per projection. Each selection produces a convex region, the intersection of four half-spaces. A bounded intersection closes on itself to form a finite solid; an unbounded one extends to infinity and has no geometric identity. The natural question is: which of these sixteen regions are bounded? The answer turns on a single algebraic identity among the four projections.

In three spatial dimensions, four unit vectors with equal pairwise angles admit no other configuration than the face normals of a regular tetrahedron, and face normals of a regular tetrahedron sum to zero. The four projections therefore satisfy

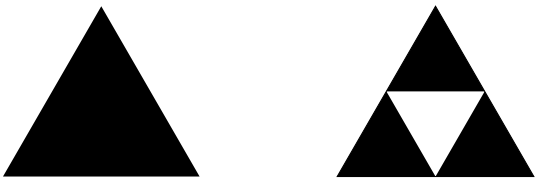
$$v_1 + v_2 + v_3 + v_4 = 0. \tag{4}$$

This identity holds at every point in coordinate space and is the sole algebraic relation among the four projections. When the inward-pointing normals of a set of bounding planes sum to zero, the origin sits at their centroid, strictly inside their convex hull, and the walls close from every direction; when they do not, a gap opens and the region escapes to infinity. For TetA, the four inward normals are the derived tetrahedral directions; they sum to zero, so TetA is bounded. This is unsurprising; TetA was defined by bounding all four projections.

But the same argument applies to the four negated directions: they also sum to zero, placing the origin at their centroid. A second bounded region exists, one that was never defined or constructed. This is TetB:

$$\textbf{TetB}: \quad \max(-v_1, -v_2, -v_3, -v_4) \leq 1. \tag{5}$$

No other region can be bounded. Consider any mixed selection, say three directions that keep their original orientation and one that is flipped. The linear relation means the flipped direction equals the sum of the other three: it is a positive combination of them. All four selected normals therefore point into the same half of space, the origin lies outside their convex hull, and the region escapes to infinity. The same reasoning applies to every other mixed selection: flipping any strict subset of the directions breaks the surrounding property, and the intersection is unbounded. Of the sixteen regions, exactly two close into finite solids: in Figure 1, the blue region is TetA and the red region is TetB. The orientation-reversed tetrahedral sector was not introduced as a separate frame; it emerges as the unique second



(a) (b)

Figure 2. The face subdivision. (a) One face of TetA, undivided. (b) The opposing vertex of TetB penetrates the face, carving it into three exposed corners and one central inverted core. This $3+1$ pattern is the first Sierpinski iteration and repeats on all four faces, producing the $4 \times 4 = 16$ boundary channels of Eq. (6).

bounded region among the sixteen half-space combinations of a single orthonormal tetrad.

The two tetrahedra interpenetrate: each vertex of TetB passes through a face of TetA, and each vertex of TetA passes through a face of TetB. Their intersection is the set of points satisfying $|v_i| \leq 1$ for all i. This region has six vertices, each obtained by setting one $|v_i| = 1$ and solving for the remaining three under the constraint $\sum v_i = 0$; all twelve edges are equal in length. The intersection is therefore a regular octahedron, visible as the gold core at the centre of Figure 1. The union of the two tetrahedra is a stellated octahedron, first described by Kepler [6] and known in modern polyhedral geometry as the stella octangula [7], here derived from the half-space inequalities of a single orthonormal tetrad.

This interpenetration couples all four faces of each tetrahedron. On each of the four coupled faces, the opposing vertex carves the face into four distinct sub-regions: three exposed triangular corners and one central inverted core. The resulting pattern is the first level of a Sierpinski subdivision, visible on every face in Figure 1. The total number of independent boundary channels across the four coupled faces follows immediately:

$$4 \text{ faces} \times 4 \text{ sub-regions} = 16 \text{ channels}. \qquad (6)$$

Each channel is a distinct connected region of the boundary surface, separated from its neighbours by the edges where TetA and TetB intersect. Figure 2 illustrates the subdivision for a single face; the numerical rendering at four million sample points confirms the full count. Whether deeper levels of the Sierpinski recursion carry geometric significance is not addressed here.

The three regions visible in Figure 1 are now fully accounted for. The blue and red tetrahedra are the only bounded regions among the sixteen half-space combinations; the gold octahedron is their intersection; and the composite stellated octahedron, with its sixteen boundary channels, is generated by the half-space inequalities of a single orthonormal tetrad, with nothing introduced and nothing added.

Structural Inventory

Every structural constant of the stellated octahedron that the tetrad produces reduces to two integers: the number of faces per tetrahedron, $K = 4$, and the spatial dimension, $D = 3$. In combinatorial topology, the standard inventory of a polyhedron is its f-vector (f_0, f_1, f_2), recording the number of vertices, edges, and faces. This count is a coordinate-independent combinatorial inventory.

Quantity	Expression	Value
Tetrahedra (TetA, TetB)	2	2
Faces per tetrahedron	K	4
Vertices (total)	2^D	8
Edges (total)	$K \times D$	12
Faces (total)	$2K$	8
Boundary channels	K^2	16
f-vector sum $(V+E+F)$	$K(K+D)$	28

The f-vector of the stellated octahedron is $(8, 12, 8)$, giving a total element count of $V + E + F = 28$. The alternating sum $V - E + F = 4$ equals twice the Euler characteristic of a sphere, consistent with two simplicially disjoint closed surfaces. Every entry in the table is determined by K and D alone. No additional parameters or geometric inputs are required.

Discussion

The preceding sections derived both tetrahedra, their interpenetration, and all sixteen boundary channels from a single input: the four half-space inequalities of one orthonormal tetrad. Of the sixteen sign combinations, exactly two are bounded, and the dual frame emerges as an analytic consequence of the primary frame's four planes. Each tetrahedron individually carries a $3+1$ face decomposition: one face whose normal has uniform sign and three faces with mixed signs, mirroring the $(-,+,+,+)$ Lorentz signature from which the construction began.

The introduction noted that the tetrad is, algebraically, the square root of the metric. Because the metric depends quadratically on the tetrad, parity-related spatial frame realisations produce the same $g_{\mu\nu}$. The two bounded regions derived above are the coordinate-space realisation of that doubled frame structure: TetA and TetB are distinct tetrad-level realisations of the same metric.

The geometry reported here is not contingent on a particular gravitational action or field equation. The two interpenetrating tetrahedra, the octahedral intersection, and the

sixteen boundary channels are consequences of the tetrad's combinatorial structure (four derived directions in three spatial dimensions) and are present before any dynamics are imposed. This geometry is consequently present in the local frame of any covariant theory formulated in the tetrad variables.

Data and Reproducibility

The point-cloud visualisation in Figure 1 was generated by evaluating the four tetrahedral half-space inequalities at four million uniformly sampled points. The rendering code is provided as an interactive HTML file (`fig1_tetrad_frame.html`) included as supplementary material. A self-contained validation script (`run_all_validations.py`) independently verifies every geometric claim in this paper, requiring only NumPy and SciPy. A derivation of the tetrahedral frame from the Lorentz signature is provided as a separate supplementary note (`supplementary_signature_to_frame.pdf`). An independent derivation of the two bounded regions from the quadratic structure of the tetrad equation is provided as a second supplementary note (`supplementary_scheme_proof.pdf`).

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